intelligence at scale. Several recent papers have endeavored to create and evaluate LLM-based forecasters (Halawi et al., 2024; Karger et al., 2024; Schoenegger et al., 2024; Turtel et al., 2025; Schoenegger et al., 2025). The rapid advances of LLMs on reasoning tasks (e.g., achieving gold medal performance at the International Math Olympiad (Google DeepMind, 2025)) offer reasons for optimism, and prompt the obvious question:

Can LLMs match or even exceed the performance of expert human forecasters?

Contributions. We answer this question in the affirmative by introducing the AIA Forecaster. On ForecastBench (Karger et al., 2024), the most widely used academic benchmark for forecasting, we achieve results that are statistically indistinguishable from human superforecasters. We also introduce a new benchmark which consists of 1610 questions sourced from a popular public prediction market platform. These questions cover decision-relevant events sourced from liquid markets, and are significantly harder than those on ForecastBench. On this benchmark, the AIA Forecaster lags slightly behind market consensus. However, an ensemble of the system's forecasts with the market consensus outperforms consensus alone, indicating that the AIA Forecaster is diversifying with respect to market prices. We corroborate these results by establishing a track record of real-time, forward-looking forecasts. In this modest sample, the AIA Forecaster is highly competitive with market consensus in liquid prediction markets.

In addition to our core experiments, we also conduct extensive analysis to understand what makes a good forecaster. Our methodological contributions are as follows:

1. The AIA Forecaster is equipped with the ability to perform a robust search over a large corpus of high-quality documents. Critically, this search is both *agentic* and *adaptive* in nature: the model is granted full discretion over what to search, and can adaptively seek out new information by conditioning on the results of earlier queries. We show there is a deep relationship between search quality and forecast quality, resolving a major open question in the literature.

2. We conduct an exhaustive analysis of *foreknowledge bias*, or unintended violations of the intended information cutoff. In particular, ex-post evaluation of such systems requires the ability to "travel back in time" and feed the forecaster news "as of" the intended forecasting date. Reliably enforcing this information cutoff is a surprisingly challenging problem (Paleka et al., 2025). We build a custom pipeline for detecting foreknowledge bias and demonstrate its efficacy, confirming the integrity of our results. We also introduce a live benchmark, MarketNightly, which includes only active, liquid prediction markets, and thus precludes the use of foreknowledge. Results on this benchmark corroborate our other findings.

3. We show that individual forecast runs are unstable, and that ensembling over many forecasts is absolutely essential to obtain frontier-level forecasting performance. We

also show that the exact method of ensembling is crucial. It is not enough to take the simple mean of forecasts. Instead, the AIA Forecaster passes the results of individual forecasting agents to a supervisor agent, which performs additional search to correct mistakes and reconcile differing perspectives.

4. LLMs are fundamentally miscalibrated for probabilistic prediction under uncertainty. They hedge too much and are too cautious. We examine a variety of different statistical correction techniques that can be applied to correct these biases. We derive a mathematical connection, which has not yet been documented in the literature, between Platt scaling and the extremization of forecasts. We also show that simple extremization techniques are often sufficient to obtain large gains in forecasting performance.

Organization. In Section 2, we provide an overview of related work. In Section 3, we describe the architecture of the AIA Forecaster, and in Section 4 we present our main results on multiple variants of ForecastBench and our novel MarketLiquid benchmark. In Section 5 we investigate the role of search, and address methodological challenges related to *foreknowledge bias*. We complement these results in Section 6, where we study real-time forecasting performance in live prediction markets. In Section 7, we study a range of ensembling and statistical calibration techniques, which we show can substantially improve on the performance of a single forecasting agent. Finally, in Section 8, we conclude with a discussion of limitations, open questions, and future work.

2 Related Work

We focus on *judgmental forecasting*, in which a forecaster is tasked with producing the probability that a binary event will occur (Lahiri and Yang, 2013; Tetlock and Gardner, 2016; Tetlock, 2017; Hanson, 1999). In the past 30 years, judgmental forecasting has become an increasingly important area of study (Lawrence et al., 2006). Two major findings that have increased its salience are (1) that the *wisdom of crowds* (aggregated predictions across many forecasters) often beats experts in judgmental forecasting (Surowiecki, 2004; Bassamboo et al., 2015; Da and Huang, 2019; Lichtendahl Jr et al., 2013), and (2) the existence of *superforecasters*, who consistently outperform even the wisdom of crowds (Tetlock, 2017; Tetlock and Gardner, 2016). The literature demonstrates that superforecasters are rare, and even subject-matter experts are often poor forecasters in their area of expertise. What's more, superforecasting ability in one domain does not necessarily transfer to other domains.

With the advent of strong AI systems, there has been increased interest in AI forecasting (Halawi et al., 2024). AI forecasting is highly scalable, so if the performance of AI forecasters matches or exceeds superforecasters, these systems can be widely deployed. Of course, in adversarial settings (e.g., financial markets), widely available AI forecasters

may in turn increase the difficulty of subsequent prediction tasks, requiring systems which continuously evolve.[1]

Techniques for AI forecasters. Modern AI forecasters provide context to an LLM and prompt it to produce the probability that a future event will occur (Wang et al., 2024; Halawi et al., 2024; Schoenegger et al., 2024). However, as with any modern AI system, there are many research choices that dramatically affect performance. For example, the LLM's prompt (Schoenegger et al., 2025) and the news provided to the LLM (Wang et al., 2024) have both been shown to carry an outsize impact on AI's forecasting ability (Turtel et al., 2025).

Benchmarks for AI forecasters. Recently, several benchmarks have been developed to evaluate the prediction quality of AI forecasters. Broadly, all benchmarks can be partitioned into two categories: *static* and *dynamic*. Static benchmarks are gathered at a single point in time and held fixed indefinitely thereafter. Dynamic benchmarks are continually updated to include fresh questions on a regular basis.

Examples of static benchmarks include Halawi et al. (2024), along with many others (Jin et al., 2020; Mutschlechner and Jatowt, 2025; Yuan et al., 2025; Wang et al., 2025; Ye et al., 2024; Zou et al., 2022). Static benchmarks will always atrophy as time marches on and new LLMs are trained. This is because newer models have knowledge of the world which is acquired after the question was initially posed. For example, if a benchmark published in 2023 asks the LLM to forecast who will win the 2024 US presidential election, models trained at later dates may acquire knowledge of the outcome (and other closely related events) through their pre-training corpus (Sarkar and Vafa, 2024). Models may also be given access to tools which inadvertently leak knowledge beyond the intended information cutoff (Paleka et al., 2025). The foreknowledge bias problem, together with several other issues related to the quality of baseline forecasts[2], is often the limiting factor in using static benchmarks to evaluate forecasting systems.

Regarding dynamic benchmarks, there are three major choices at present: ForecastBench (Karger et al., 2024), Prophet Arena (Tao et al., 2025), and the Metaculus AI Forecasting Benchmarking Series (Metaculus, 2025). ForecastBench sources questions from public prediction markets, including Manifold, Polymarket, and Metaculus. It filters out "low-quality" questions using an LLM. Given a list of questions, it then produces "combination questions", which ask the model to forecast boolean conjunctions of the original events. Prophet Arena also sources questions from prediction markets, but fixes the news sources the LLMs can use. This simplifies the forecasting task significantly because, as we argue in Section 5, effective search is arguably the most critical component of effective forecasting. The Metaculus AI Forecasting Benchmarking Series releases

1. More generally, tasks in which predictions influence the distribution of future outcomes are known as *performative* prediction tasks (Perdomo et al., 2020).
2. In particular, if one does not collect high-quality superforecaster or crowd predictions at the time the benchmark is created, it is impossible to travel back in time and obtain these baseline forecasts. Many extant benchmarks lack such baselines, which are crucial for evaluation.